

Generative Adversarial Nets

DATA 443/643 · Week 11, note 11.1

Week 11

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Ideas from Stanford CS231N 2025 L13 (give up on $p(x)$; sample in one pass). Original slides; that course is not copied.

Lecture plan

- 1 G maps z to a fake. D scores real vs fake.
- 2 No likelihood, only a sampler
- 3 Bayes D when G parks on one spike
- 4 One pretty sample is not a trained G

Key goal: G is a distribution. D is a moving classifier.

First time: what / why / formula

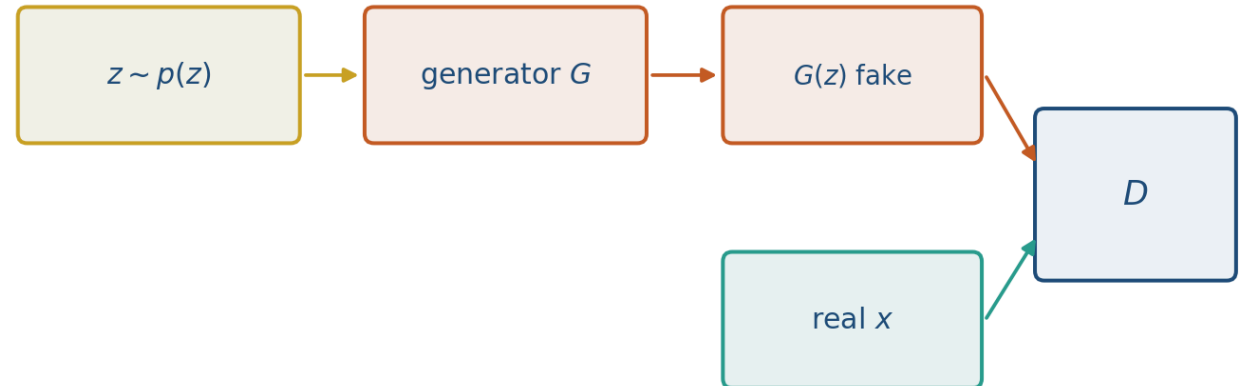
- What / why. You give up on writing $p(x)$. $G(z)$ maps noise to a fake. $D(x) \in (0, 1)$ says “real.” Likelihood models can be blurry or slow to sample. GANs aim for sharp one-pass samples (images, later 11.3’s coverage bugs).
- Architecture. $z \sim p(z) \rightarrow G \rightarrow \text{fake}$. D sees reals from the data and fakes from G . No encoder required.
- Formula. $D^*(x) = p_{\text{data}}(x) / (p_{\text{data}}(x) + p_G(x))$. At a perfect G , $D^* = 1 / 2$.
- Tradeoffs. + Fast sampling, sharp samples when it works. – No likelihood, unstable train (11.2), mode collapse (11.3), not an LLM method.

How. Train both (details next note). Bayes-optimal D is $1 / 2$ where G matches the data.

Two players, one sample

- $z \sim \mathcal{N}(0, I)$. $x = G(z)$.
- D never sees z . G never sees a real x on its own update.
- Implicit density, direct sample.

GAN: G tries to fool D ; D tries not to be fooled



The optimal discriminator

$$D^*(x) = \frac{p_r(x)}{p_r(x) + p_g(x)}$$

If $p_g = p_r$, $D^* = 1 / 2$ on every $\mathcal{X}$.

If G never visits a mode, D is 1 there and G never hears about it.

You cannot read $p(x)$ off a trained GAN.

Two spikes, G parked on +2

G parked on +2. Bayes $D(+2) = 1/2$; the left mode is uncontested.

- Reals: half at -2 , half at $+2$.
- Bayes $D(+2) = 0.5$. $D(-2) = 1$.
- A coin flip at $+2$ is not coverage of the mixture.

location	what D sees	$D(x)$
$x = +2$	half real, half fake	0.5
$x = -2$	only reals	1
elsewhere	only fakes, if any	≈ 0

What you can and cannot do

A GAN

Draw a fake in one forward pass. No $p(x)$, no $p(z|x)$, no test log-likelihood.

Week 12 diffusion

An explicit noise schedule. T reverse steps. You still sample; you also have a regression loss.

What to take from this lecture

Sampler

G is not an autoencoder. The only teacher is D's scalar.

Moving D

Yesterday's perfect D scores a different G today.

Lab 11

8-D noise → 2-D point. Two Gaussians, not faces.